

TEEPARTHI SATYA SAI NATH

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Summary

M.Tech Computer Science student specializing in Machine Learning, Deep Learning, and Agentic AI systems. Experience building multi-agent LLM frameworks, transformer-based security systems, and predictive ML pipelines. Proficient in Python, SQL, TensorFlow, PyTorch, Scikit-learn, and LLM orchestration frameworks (LangChain-style agents, RAG, MCP). Seeking roles in Machine Learning Engineering, Data Science, AI/Agentic Systems Engineering, and Software Development.

Education

Master of Technology in Computer Science and Engineering — Jul 2025 – Present
Amrita School of Computing, Bengaluru — CGPA: 8.22/10.00

Bachelor of Technology in Computer Science and Engineering — Dec 2021 – May 2025
Parul Institute of Engineering and Technology, Vadodara — CGPA: 7.47/10.00

Projects

OmniBug: Agentic Multi-Language Bug Detection Framework — 2026

- Engineered a cascading dual-agent LLM pipeline (Detector + Describer agents) that localizes, categorizes, and explains bugs across 9+ languages including Python, Java, JavaScript, TypeScript, Go, Rust, C, C++, SQL, and the domain-specific language SmartRDI.
- Implemented Retrieval-Augmented Generation via a custom FastMCP server to inject domain-specific documentation into agent context, improving detection accuracy on niche APIs and hardware-constrained code.
- Designed a cascading model-fallback strategy (fast low-latency models with automatic escalation to larger models on rate limits) to balance throughput and analysis depth; classified issues into 10 structured bug categories (concurrency errors, null references, resource leaks, API misuse, and more) with results exported to CSV reports.

PromptGuard: Hybrid Transformer-Based Prompt Injection Detection System — 2026

- Designed a real-time prompt injection detection framework for LLMs using a hybrid architecture combining a fine-tuned DistilBERT transformer encoder with rule-based regex heuristics, trained on approximately 43K samples aggregated from six public datasets.
- Achieved 0.93 macro F1 score with low false-positive rates, enabling high-precision classification of malicious prompts as a deployable API layer for LLM safety pipelines.
- Applied NLP preprocessing, adversarial prompt augmentation, and threshold calibration to improve detection robustness across diverse attack patterns.

Internship Experience

Software & Machine Learning Intern — Jan 2025 – Apr 2025
Edu-Versity, AMG Technologies

- Built and deployed predictive ML models using Python and Scikit-learn, incorporating feature engineering, data preprocessing, and cross-validation pipelines.
- Reduced model error through iterative hyperparameter tuning and ensemble techniques; documented results using structured evaluation metrics (RMSE, F1-score, AUC).

Technical Skills

Languages	Python, SQL, JavaScript (Basic)
Machine Learning / AI	Deep Learning, NLP, Supervised Learning, Model Evaluation, Hyperparameter Tuning, Scikit-learn, TensorFlow, PyTorch
Agentic AI / LLM Systems	Multi-Agent Architectures, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), LlamaIndex, HuggingFace Transformers, Prompt Engineering, Google Gemini API, Groq API
Data & Analytics	Data Cleaning, EDA, Feature Engineering
Software Engineering	OOP, SDLC, Algorithm Optimization, Operating Systems (Basic), Computer Networks (Basic)
Testing & QA	Unit Testing, Debugging, Model Validation, Cross-Validation
Tools & Platforms	Git, JupyterLab, VS Code, Linux

Certifications

NPTEL Certified in Software Engineering